Executive Summary: Resilient Cross-Asset Portfolio

Problem and approach

Most portfolios optimise for normal conditions. This project asks which assets remain acceptable under a named Taiwan Strait / TSMC disruption.

The pipeline predicts 12-month returns, re-predicts them under the shock, then allocates using baseline return, stressed return and stressed downside.

Main findings

- 104 instruments across stocks, country ETFs and commodities.
- Latest cache uses live yfinance prices and fundamentals for all 104 assets.
- Split-conformal calibration gives 90.1% validation coverage for the nominal 90% interval; test coverage is 81.5%.
- Resilient allocation earns less in the realised test window, but improves the Taiwan-scenario 5% tail forecast from -21.7% to -16.2%.
- SHAP confirms the stress mainly acts through volatility, drawdown and political-stability context.

Recommendation

Use the resilient allocation as a stress-aware overlay, not as a claim of unconditional benchmark outperformance. Tune alpha for normal-versus-shock growth and lambda for downside aversion.

Highlights

- Reproducible end-to-end data, model, scenario, allocation and explanation pipeline.
- Uses ensembles, prediction uncertainty, counterfactual scenarios and SHAP.
- Concrete disruption objective rather than a generic risk score.
- Includes sensitivity checks for the allocation knobs.

Limitations

- No actual Taiwan shock occurs in the test window, so realised performance and scenario resilience are different objectives.
- Test coverage remains below 90% after conformal calibration.
- Scenario magnitudes are stress assumptions, not identified causal effects.